

# TMX-090 closure evidence — I1 WHEEL-COPY-MODE- OVERRIDE-001

**Date:** 2026-09-29 **Investigator:** Claude (systematic-debugging arc, root-cause investigation per §11.4.102)

## Summary

Issues.md item I1/TMX-090 was tracked as "In progress — cause NOT established" for test 17 sub-check T3 ("FAIL: T3: live WheelUpPane binding is not the copy-mode override"). Investigation confirms the underlying defect was **already root-caused and already fixed** in commit 2e4ec22 (2026-09-01 18:27:50 +0200, "fix(tests): version-stable key readback + six skip remediations (TMX-090/091)") — but the Issues.md entry was added by that SAME commit, describing the investigation's pre-diagnosis state (before the root cause was found later in that work session), and was never updated afterward. This closure corrects that tracker-sync gap.

## Root cause (proven in 2e4ec22's own commit message, independently re-confirmed

live today)

tmux 3.7b's `cmd-list-keys.c` print branch is:

```
if ((single && tc != NULL) || n == 1)
```

The `|| n == 1` arm has NO `tc != NULL` guard, so ANY `list-keys` query matching exactly one row is routed to a nonexistent client's status line and DISCARDED when run from a script (no attached client). The raw single-key query form, `tmux list-keys -T root WheelUpPane`, therefore returns `rc=0` with EMPTY output even though the binding is genuinely present and correct — a §11.4.201(6) false null, not a missing binding.

## Fix (already landed, 2026-09-01, commit 2e4ec22)

`scripts/tests/lib/list_key.sh` — a version-stable `tmx_list_key` helper that lists the WHOLE key table and field-matches via `awk`, rather than querying a single key. Wired into 10 call sites across tests 17/44/46/47/48, replacing the raw `list-keys -T <table> <key>` form. A paired mutation, `M-LIST-KEY-VERSION-STABLE` (`scripts/tests/meta_test_false_positive_proof.sh`), reverts the helper to the broken single-key form and asserts test 17's T3 goes FAIL — proving the regression guard is real per §1.1.

## Independent re-verification performed today (2026-09-29), on a LIVE fresh build

Two independent subagents, two independent fresh builds (`bash scripts/setup.sh --rebuild`), both confirming `tmux 3.7b`:

- 1. Direct capture (test 17):** standalone run of `scripts/tests/17_scrollback_copy_mode.sh` — `PASS=13 FAIL=0 SKIP=0, EXIT=0`, including explicit `PASS: T3: live WheelUpPane binding drives copy-mode scroll-up (override active, not tmux default)`. Same result reproduced inside the full `setup.sh` verification-gate run (not a standalone-only fluke). The raw single-key form (`list-keys -T root WheelUpPane`) was independently re-queried live and reproduced the documented `tmux 3.7b` quirk exactly: `rc=0`, empty output. The full `list-keys -T root table` (27 rows) shows `WheelUpPane` present exactly once, correctly formed, matching `scripts/tmux.conf.template:81` verbatim — no double-binding, no stale config, config file confirmed as the tracked repo path (`/home/milosvasic/Projects/tmux/scripts/tmux.conf.template`, not a stale system-wide copy).
- 2. Sibling check (test 47, same mechanism):** standalone run of `scripts/tests/47_alt_screen_scroll.sh`, twice, deterministic — `PASS=8 FAIL=0 SKIP=0, EXIT=0` both times, including `PASS: T6: live WheelUpPane binding overrides tmux default — drives copy-mode unconditionally even under mouse-tracking`. T6 uses the identical `tmx_list_key` helper and assertion shape as T3.
- 3. Static trace (Phase 2 pattern analysis):** confirmed the config mechanism itself is unchanged and clean — the binding at `scripts/tmux.conf.template:81` was introduced once in commit `d0825e1c7`

(2026-05-21) and never modified since; tmx (both Linux and Darwin branches) loads this exact file directly via -f/source-file on every new/attach/ reload, with no templating/sed touching the line and no OS-conditional exclusion; the binding syntax is confirmed valid for the pinned tmux 3.7b by both the shipped man page and tmux's own compiled-in default wheelUpPane binding using the identical bind -n WheelUpPane if -F ... grammar form.

4. **Meta-test mutation re-verification (today, this closure):** scripts/tests/meta\_test\_false\_positive\_proof.sh (the FULL suite, including M-LIST-KEY-VERSION-STABLE) was run in full against this same fresh build to confirm the regression guard for this fix still functions. Result recorded separately in the release-gate evidence for this cycle (also satisfies CONTINUATION.md §3.37b's outstanding "whole-suite meta-test mutation sweep NOT RUN" item).

## Verdict

**FIXED.** The live product behaviour is correct and has been correct since commit 2e4ec22 landed on 2026-09-01. Today's independent re-verification on a freshly-built binary confirms no regression since. Issues.md I1/TMX-090 is closed as a §11.4.7-compliant demotion: the FAIL no longer reproduces, under the SAME conditions (same test, same live-server readback mechanism, same config), backed by captured evidence, not merely "cannot reproduce in isolation."

**Distinct finding, tracked separately, not part of this closure:** the Issues.md entry itself was never updated after its own fix landed in the same commit — a tracker-hygiene gap of the same class already found and being addressed this cycle for A2/D2 (stale Fixed status never migrated). No new item filed for this specific instance since it is fully resolved by this closure.